

Valvetrain Result Viewer

AVL Excite Timing Drive — Technical Documentation

Project	AW82001 — V8 Engine Valvetrain Dynamic Analysis
Model	vtRBint01.Ref_C10 (Intake Right Bank, Reference Config C10)
Software	AVL Excite Timing Drive
Scope	Valvetrain functionality verification up to 7 500 rpm
Tool	valvetrain_viewer.py (PySide6 + Matplotlib interactive GUI)
Date	2026-06-08

1. Overview

The Valvetrain Result Viewer is an interactive desktop application for post-processing AVL Excite Timing Drive simulation results. Its primary purpose is to verify valvetrain functionality — valve lift, Hertzian contact pressure, and spring behaviour — across the engine speed range 7 000 – 7 500 rpm for model **vtRBint01.Ref_C10**.

The tool uses a **drag-and-drop** workflow: result tiles in the left panel represent individual signal channels from the GID result files; dragging a tile onto the plot canvas instantly loads and displays the time-history curve against crank angle. Multiple tiles from different RPM points or engine components can be overlaid on the same subplot for direct comparison.

2. GUI Layout

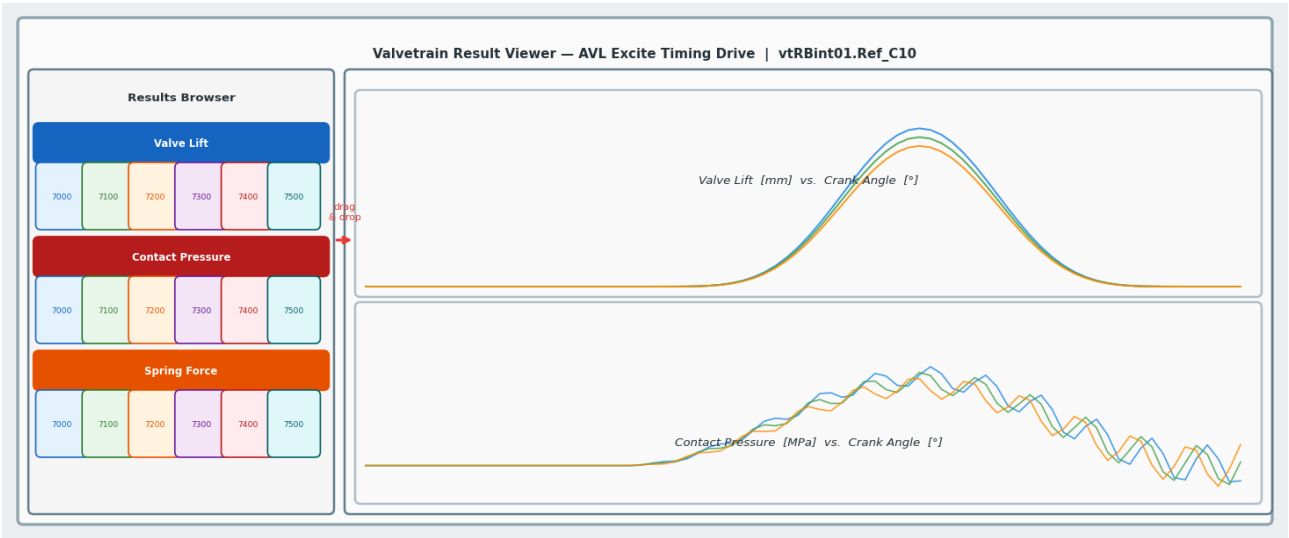


Figure 1 — Schematic of the application window. Left: Results Browser with colour-coded RPM tiles. Right: dual-subplot canvas (Valve Lift top, Contact Pressure bottom).

2.1 Results Browser (left panel)

The left panel lists all available signal channels for all six RPM operating points simultaneously. Tiles are organised hierarchically: **signal category** → **RPM sub-header** → **component tile**. Each tile is colour-coded by engine speed (see Table 1) to make it easy to identify at a glance which operating point is being added to the plot.

Engine Speed	Background colour
7 000 rpm	#E3F2FD (light blue)
7 100 rpm	#E8F5E9 (light green)
7 200 rpm	#FFF3E0 (light orange)
7 300 rpm	#F3E5F5 (light purple)
7 400 rpm	#FFE0E0 (light red)
7 500 rpm	#E0F7FA (light teal)

Table 1 — RPM colour coding of result tiles.

2.2 Plot Canvas (right panel)

The plot area contains two stacked subplots sharing the x-axis (crank angle 0 – 720°). Routing of dropped tiles is automatic: **Valve Lift** and **Valve Seat Force** tiles always appear on the *top* subplot; **Contact Pressure** and **Contact Force** tiles appear on the *bottom* subplot. Other signal categories (spring force, lash adjuster, ...) default to the top subplot. Data is extracted from the last complete 720° crank-angle cycle recorded in the GID file and normalised to the range 0 – 720°.

3. Available Signal Categories

Category	Source file prefix	Channel	Unit	Subplot
Valve Lift	VAFA_	lift	mm	Top
Valve Seat Force	VAFA_	seat force	N	Top
Contact Pressure	CLUB_	contact stress	MPa	Bottom
Contact Force	CDAT_	force	N	Bottom
Spring Force	CTOR_	force	N	Top
Spring Coil Contact	SPPR_	force coil contact	N	Top
Lash Adjuster	HLIF_	lift / force / pressure	mm / N / Pa	Top
Finger Follower	FIFO_	lift / force	mm / N	Top

Table 2 — Result categories, source file prefixes, and default subplot assignment. Units are converted automatically: m→mm for lift, N/m²→MPa for contact stress.

4. Sample Results (INTR_VAFA1 / INTR_CLUB1)

4.1 Valve Lift

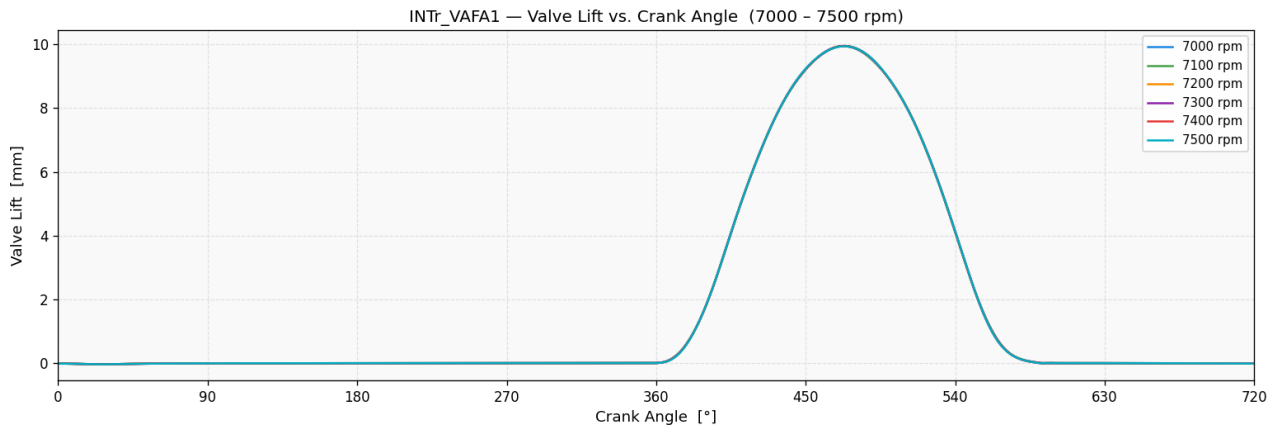


Figure 2 — Valve lift (INTr_VAFA1, intake valve 1, right bank) vs. crank angle for all six speed points 7 000 – 7 500 rpm. The curves are nearly identical, indicating stable kinematic behaviour across the speed range. Peak lift $\approx$ 9.95 mm.

4.2 Cam-Follower Contact Pressure

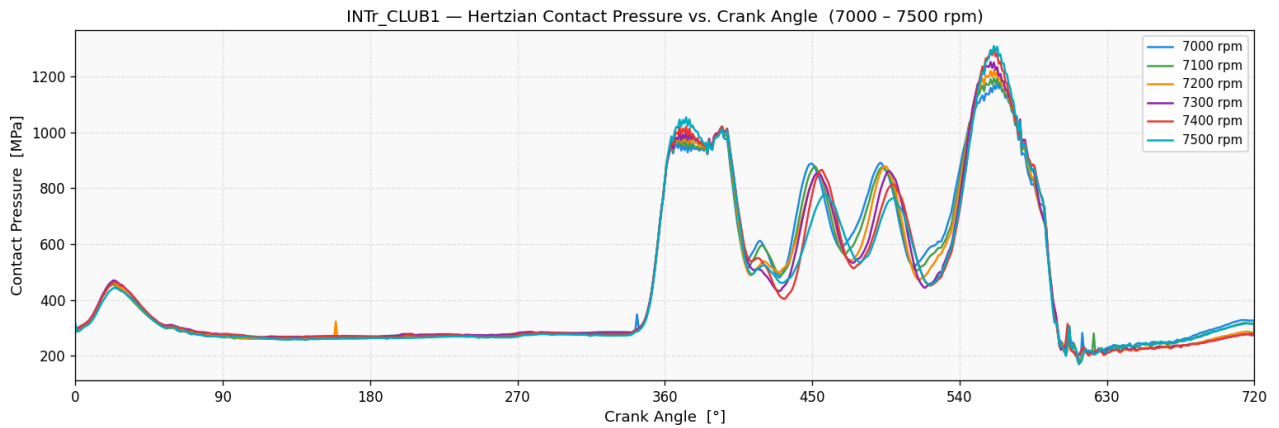


Figure 3 — Hertzian contact pressure (INTr_CLUB1, cam-follower contact, right bank) vs. crank angle. Peak pressures reach approximately 1 170 MPa in the opening/closing flanks at 7 500 rpm. Pressure oscillations during the lift event are caused by dynamic contact force variation (spring surge, follower inertia).

5. Data Structure and File Format

Results are stored in **AVL GID** text files (ASCII, Latin-1 encoding). Each file contains a structured header block followed by columnar numeric data. The header is delimited by BEGIN / END keywords and defines the simulation speed, component name (objectname), channel names, and units. Continuation lines use the & character.

Example GID header (VAFA_10.GID):

```
BEGIN
speed = '0.700000E+04'
objectname = 'INTr_VAFA1'
CHANNEL = ['time', &
'equiv. cam angle', &
'equiv. crank angle', &
'ref. angle', &
'lift', 'velocity', 'acceleration', 'force', 'seat force']
UNIT = ['s', 'deg', 'deg', 'deg', 'm', 'm/s', 'm/s^2', 'N', 'N']
END
0.034309 720.499 1440.998 720.499 -1.576e-05 -0.00690 -80.95 -179.2 3947.8
```

...

Path element	Description
excite_td/	Root result directory
vtRBint01.Ref_C10/	AllCaseSet summary (speed-sweep statistics)
vtRBint01.Ref_C10.7000rpm/results/	Time-domain results at 7 000 rpm
...	7 100 – 7 500 rpm (same structure)
VAFA_<id>.GID	Valve analysis: lift, velocity, force, seat force
CLUB_<id>.GID	Contact lubrication: Hertzian stress, OFT, friction
CDAT_<id>.GID	Cam data / contact force, friction moment
SPPR_<id>.GID	Spring coil: force, coil contact force
CTOR_<id>.GID	End-coil spring force
HLIF_<id>.GID	Hydraulic lash adjuster: lift, pressure, flow
FIFO_<id>.GID	Finger follower: lift, force

Table 3 — Directory layout of the excite_td result tree.

6. Usage Instructions

1. Launch

Open a terminal in the project directory and run:

```
python valvetrain_viewer.py
```

The application starts loading all six RPM catalogs in the background.

2. Browse tiles

The Results Browser (left panel) lists signal tiles grouped by category and RPM. Scroll down to see all eight categories. The tile colour indicates the engine speed.

3. Drag a tile

Click and hold any tile, then drag it onto the plot canvas. Release the mouse to drop. The curve appears immediately on the appropriate subplot.

4. Overlay multiple curves

Drag additional tiles from the same category but different RPM points to compare speeds. Repeat for different valve components (INTr_VAFA1 – VAFA8) to compare cylinders.

5. Navigate the plot

Use the matplotlib toolbar (zoom, pan, home, save) at the top right of the canvas.

6. Remove curves

Remove last undoes the most recently added curve. *Clear all* resets both subplots.

7. Software Dependencies

Package	Version tested	Purpose
Python	3.13+	Runtime
PySide6	6.11.0	Qt6 GUI framework (window, drag & drop)
matplotlib	3.10.9	Embedded plot canvas, navigation toolbar
numpy	2.4.4	Numeric array operations, data parsing
reportlab	any	PDF documentation generation (this document)